

MATH 4720 / MSSC 5720

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Chapter 5 (Part C)



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CHAPTER 5 (PART C)

- Confidence Interval (CI)
- CI for μ , when σ is known
- Choosing Sample Size for Estimating μ
- A Statistical Test for μ
- Type I and Type II Errors
- Power of a Test
- Choosing Sample Size for Testing μ
- Level of Significance
- P-value
- Inference about μ , when σ is unknown

LEVEL OF SIGNIFICANCE

- The approach to hypothesis testing we discussed so far is called classical approach. Note that it is based on a decision rule for a given α .
- A question arises: What is an appropriate α ?
- Many statisticians object to this approach. They say that an appropriate approach should be that we should indicate how much evidence the data provide in favor of H_0 .
- What to do?
- p – value
= $P(\text{observing the evidence more extreme than the evidence by the data if } H_0 \text{ is true})$

P-VALUE APPROACH

- **Example:**

An experiment was done on 35 petri dishes which were first cultured with E.Coli bacteria, and then a solution of the antibacterial soap was added. After 24-hour incubation period, bacterial counts were recorded on each of 35 dishes.

- **Sample Information:** $\bar{y} = 31.2, \quad s = 8.4$

- $H_0: \mu = 33$

- $H_a: \mu < 33$

- **T.S.** $z = \frac{\bar{y} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{31.2 - 33}{8.4/\sqrt{35}} = -1.27$

- [Applet: P-value \(Link #2\)](#)

P-VALUE APPROACH CONT'D

- $H_0: \mu = 33$ vs $H_a: \mu < 33$. **T.S.** $z = -1.27$

- p - value
= $P(\text{observing the evidence more extreme than the evidence by the data} \mid H_0 \text{ is true})$
= $P(Z \leq -1.27) = 0.1020$

- **Since this evidence for H_0 is not small enough, we do not have sufficient evidence to reject H_0 .**

- **Suppose $\bar{y} = 30.2$ instead of 31.2**

$$Z = \frac{\bar{y} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{30.2 - 33}{8.4/\sqrt{35}} = -1.972$$

- p - value = $P(Z \leq -1.972) = 0.0244$
- **Since this evidence for H_0 is very small, we should reject H_0 .**

ANOTHER INTERPRETATION OF P-VALUE:

- $p - value$
- = smallest level of α at which H_0 can be rejected.
- In the previous example:
- When $\bar{y} = 30.2$, $p - value = 0.0244$. This is the smallest level at which H_0 can be rejected.
- If $\alpha = 0.05$, $p - value < \alpha$, so **Reject H_0**
- If $\alpha = 0.01$, $p - value > \alpha$, so **Fail to Reject H_0**
- In other word:
 - You **Reject** at any $\alpha > 0.0244$
 - You **Fail to Reject** at any $\alpha < 0.0244$

GENERAL APPROACH

- If $p\text{-value} < \alpha$, **Reject H_0 in favor of H_a .**
- **Computing p-value**
 - $H_0: \mu = \mu_0$
 - **Compute** $z = \frac{\bar{y} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$
 - $H_a: \mu > \mu_0$ $p\text{-value} = P(Z > \text{computed } z)$
 - $H_a: \mu < \mu_0$ $p\text{-value} = P(Z < \text{computed } z)$
 - $H_a: \mu \neq \mu_0$ $p\text{-value} = 2 * P(Z > |\text{computed } z|)$
- **Applet: P-value**

THREE EXAMPLES

- **Let $\alpha = 0.05$ in the following examples.**

1. $H_0: \mu = 5.0$ vs. $H_a: \mu > 5.0$

$$n = 16, \quad \bar{y} = 6.5, \quad \sigma = 2.0$$

Given that sample is drawn from normal population.

$$- \quad Z = \frac{\bar{y} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{6.5 - 5.0}{2.0/\sqrt{16}} = 3.0$$

$$\begin{aligned} - \quad \text{p-value} &= P(Z > 3.0) \\ &= \text{normcdf}(3.0, \infty, 0, 1) \\ &= 0.0013 \end{aligned}$$

- **Conclusion: Since this p-value $< \alpha = 0.05$, we reject H_0 in favor of H_a .**

THREE EXAMPLES CONT'D

2. $H_0: \mu = 150$ vs. $H_a: \mu < 150$

$n = 50, \quad \bar{y} = 147.4, \quad \sigma = 10$

– $Z = \frac{\bar{y} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{147.4 - 150}{10/\sqrt{50}} = -1.84$

– p-value = $P(Z < -1.84)$
= $\text{normcdf}(-\infty, -1.84, 0, 1)$
= 0.0329

– **Conclusion:** Since this p-value $< \alpha = 0.05$, we reject H_0 in favor of H_a .

THREE EXAMPLES CONT'D

3. $H_0: \mu = 35$ vs. $H_a: \mu \neq 35$

$$n = 25, \bar{y} = 40, \sigma = 15$$

Given that sample is drawn from normal population.

$$- Z = \frac{\bar{y} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{40 - 35}{15/\sqrt{25}} = 1.67$$

$$\begin{aligned} - \text{p-value} &= 2 * P(Z > 1.67) \\ &= 2 * \text{normcdf}(1.67, \infty, 0, 1) \\ &= 0.095 \end{aligned}$$

- Conclusion: Since p-value $\nless 0.05$.

We don't have sufficient evidence to reject H_0 in favor of H_a .

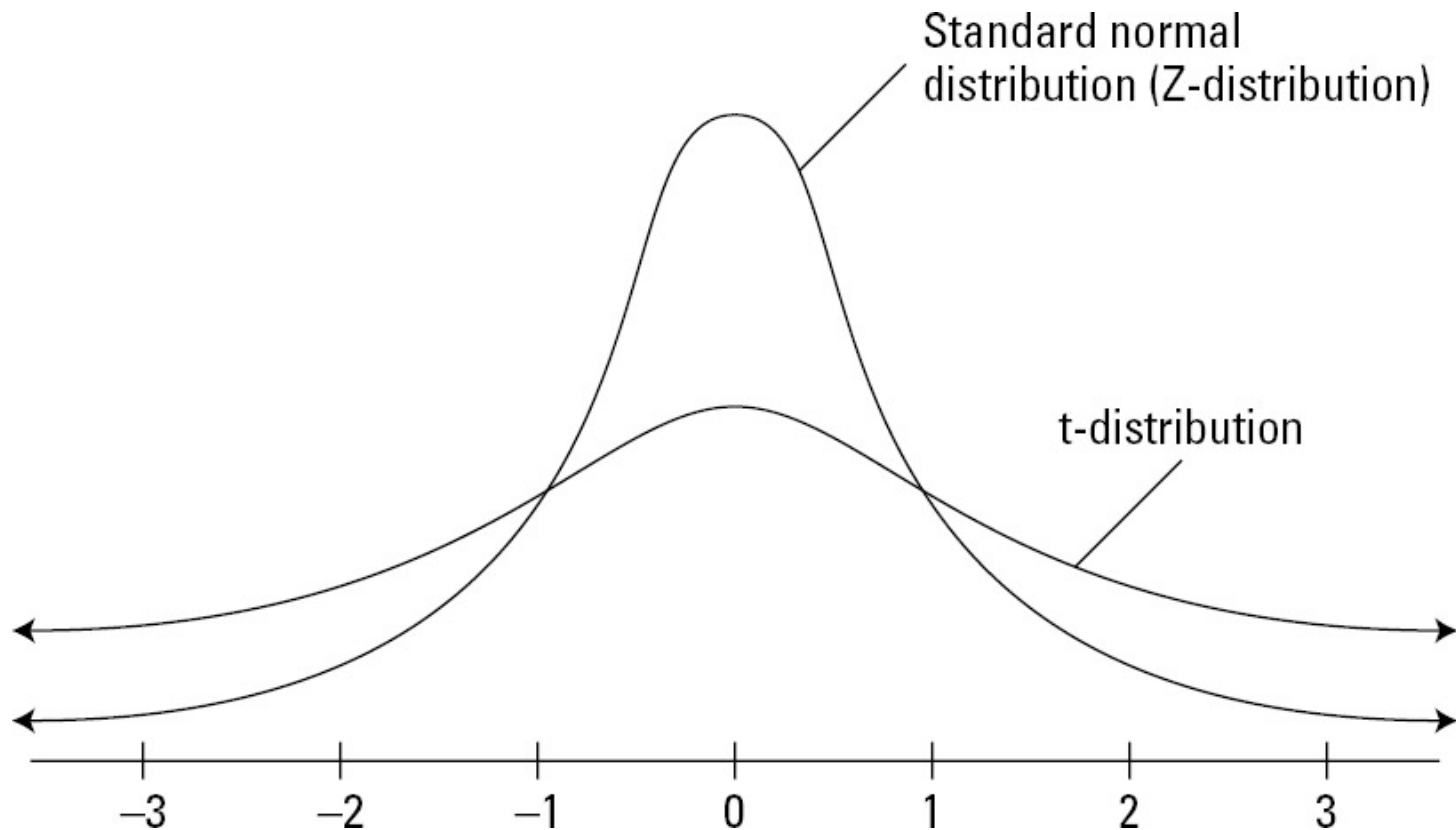
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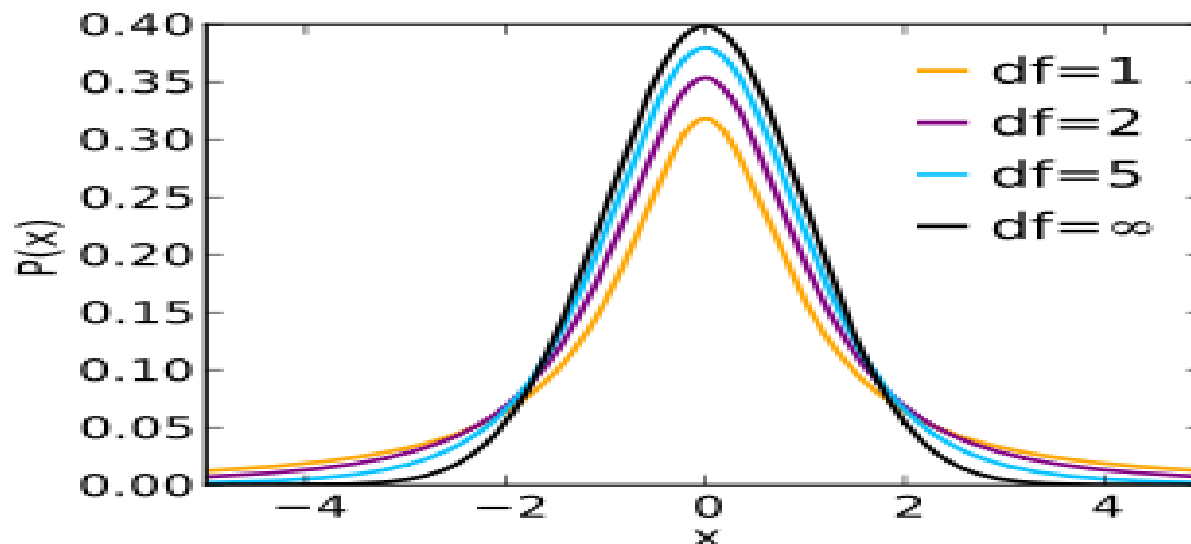
STUDENT T-DISTRIBUTION

- $$t = \frac{\bar{y} - \mu}{\frac{s}{\sqrt{n}}}$$
- **The distribution of t is called student-t distribution.**



REMARKS ON T-DISTRIBUTION

1. **t-dist. has similar shape as $N(0, 1)$, but is flatter than $N(0, 1)$.**
2. **The t-distribution is symmetric around 0 as $N(0, 1)$ is.**
3. **It has a range from $-\infty$ to ∞ as the range of $N(0, 1)$.**
4. **Unlike $N(0, 1)$, the t-distribution depends on the **degrees of freedom** df.**
5. **As the $df \rightarrow \infty$, t-distribution approaches to $N(0, 1)$.**



- [Applet: t-distribution vs \$N\(0,1\)\$](#)

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EXAMPLE

- **Consider a population of hypertension group whose average systolic blood pressure (SBP) is 150. You want to determine whether a new treatment is effective in reducing SBP. A clinical trial was conducted on 25 patients of this population. After 6 months of treatment, SBP was recorded on each subject.**
- $\bar{y} = 147.2, \quad s = 5.5$
- **Is there a sufficient evidence at $\alpha = 0.05$ that the new treatment is effective?**
- **Assume that the distribution of SBP is normal.**

EXAMPLE CONT'D

- $H_0: \mu = 150$ vs. $H_a: \mu < 150$
- **TS:** $t = \frac{\bar{y} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{147.2 - 150}{\frac{5.5}{\sqrt{25}}} = -2.55$
- **Decision Rule:** Reject H_0 in favor of H_a if $t < -t_\alpha$
- $df = n - 1 = 25 - 1 = 24$, $\alpha = 0.05$
- $t_\alpha = \text{invT}(0.95, 24) = 1.711$
- **Reject H_0 in favor of H_a if $t < -1.711$.**
- **Conclusion:** Is $t < -1.711$? Yes, since $t = -2.55$.
Thus we reject H_0 , and **we have sufficient evidence to conclude that the new treatment is effective.**

P–VALUE APPROACH

- $H_0: \mu = 150$ vs. $H_a: \mu < 150$, **TS:** $t = -2.55$
- p–value = $P(t < -2.55)$
= $\text{tcdf}(-\infty, -2.55, 24) = \mathbf{0.0088}$
- **Since p–value $< \alpha = 0.05$, we reject H_0 in favor of H_a .
We have sufficient evidence to conclude that the new treatment is effective.**
- **P–value Formula:**
 - $H_a: \mu > \mu_0$, p–value = $P(t > \text{computed } t)$
 - $H_a: \mu < \mu_0$, p–value = $P(t < \text{computed } t)$
 - $H_a: \mu \neq \mu_0$: p–value = $2 * P(t > |\text{computed } t|)$

ESTIMATION OF μ USING A CONFIDENCE INTERVAL

- $100(1 - \alpha)\%$ **Confidence Interval of μ is**

$$\bar{y} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

- **Assumption: Either**
 - $n \geq 30$
 - **the sample is drawn from a normal population.**

BOOK EXERCISE 5.39

- **A discount tire manufacturer claim that its tires can be driven at least 35,000 miles on the average. A consumer testing agency suspects that this claim is false. A study was conducted on 15 cars in which the testing tires were mounted.**

$$n = 15, \bar{y} = 31.47, s = 5.04 \text{ (in thousand miles)}$$

- (a) Estimate the mean miles driven by the tires using 99% confidence interval.**
- (b) Is there a sufficient evidence ($\alpha = 0.01$) that the manufacturer's claim is false.**

FIRST LET'S DO (B)

- $n = 15, \bar{y} = 31.47, s = 5.04$
- **(b)** $H_0: \mu = 35$ vs. $H_a: \mu < 35$
- **TS:** $t = \frac{\bar{y} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{31.47 - 35}{\frac{5.04}{\sqrt{15}}} = -2.71$
- **Decision Rule:** $\alpha = 0.01. df = 15 - 1 = 14, t_\alpha = 2.624$
- **Reject H_0 in favor of H_a if $t < -t_\alpha = -2.624$**
- **Conclusion: Is $t < -2.624$? Yes. Reject H_0 in favor of H_a .**
- **We have sufficient evidence to conclude that the miles driven by the tires is less than 35,000 miles on the average.**

NOW LET'S DO (A)

- $n = 15, \bar{y} = 31.47, s = 5.04$

(a) 99% confidence interval of μ .

- **Formula:** $\bar{y} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$
- **Assumption:** sample is drawn from a normal population
- $\alpha = 0.01, \alpha/2 = 0.005, df = 14, t_{\alpha/2} = 2.977$
- 99% **CI of μ :** $31.47 \pm 2.977 * \frac{5.04}{\sqrt{15}}$
 31.47 ± 3.90
- $27.59 < \mu < 35.3$

SUMMARY



Case 1: One Variable

	Numerical Variable, σ known	Numerical Variable, σ unknown
Parameter of Interest:	Mean, μ	Mean, μ
Confidence Interval Formula:	$\bar{x} \pm z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$	$\bar{x} \pm t_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$ with $df = n - 1$
Name of Hypothesis Test, H_0	One Sample Z Test, $H_0 : \mu = \mu_0$	One Sample T Test, $H_0 : \mu = \mu_0$
Test Statistic Formula:	$z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$	$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$ with $df = n - 1$
p-value:	$H_a : \mu \neq \mu_0$, p-value = $2P(Z \geq z)$ $H_a : \mu > \mu_0$, p-value = $P(Z \geq z)$ $H_a : \mu < \mu_0$, p-value = $P(Z \leq z)$	$H_a : \mu \neq \mu_0$, p-value = $2P(T \geq t)$ $H_a : \mu > \mu_0$, p-value = $P(T \geq t)$ $H_a : \mu < \mu_0$, p-value = $P(T \leq t)$